Weighted Cycles and Finite-Resolution Operators on an Explicit Hyperbolic Hénon Survivor

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Abstract

We study weighted periodic-orbit data and finite-volume transfer matrices for the reversible, area-preserving Hénon map $H_6(x, y) = (1 - 6x^2 - y, x)$. Four closed rational state rectangles support a nonempty, locally maximal, uniformly hyperbolic survivor, while the numerical finite-volume matrices use their strict open interiors with absorbing escape. A signed-square-root contraction and a two-sided cone argument give a conjugacy with a mixing four-state subshift, topological entropy $\log((1 + \sqrt{5})/2)$, and an exact bijection between primitive symbolic necklaces and primitive survivor orbits. Consequently, the 79 primitive necklaces through period twelve are an exact orbit count; high-precision refinement supplies their coordinates and multipliers. The unweighted cycle coefficients reproduce the finite-state determinant, while the multiplier-weighted Euler determinant and the periodic-point flat determinant provide an algebraically related truncation-consistency check. Independently assembled open finite-volume matrices give six finest-grid leading moduli within 1.22% of the period-twelve cycle reference. A paired fine-to-coarse projection experiment passes its gap component: all eight direct/common-cloud group medians and their pooled median are below 2%. Only two of four chain-by-budget groups pass the absolute dyadic-contrast criterion, however, so the aggregate dyadic component and overall G1 fail. The inherited projection budget prevents a causal attribution to common sampling. Moreover, a preregistered localization variable degenerates to target occupancy; after conditioning on occupied rows, no robust internal-cell phase association remains. The results establish an explicit hyperbolic testbed and an auditable finite-resolution bridge, without asserting continuous-operator convergence or a global Hénon zeta identity.

1 Introduction

Periodic orbits and transfer-operator resonances describe the same open chaotic system from two different computational directions. In one direction, an orbit catalogue supports Artin–Mazur or thermodynamic zeta functions [Artin and Mazur, 1965, Ruelle, 1976, 1978]. In the other, finite-volume or Ulam matrices approximate the action of a Perron–Frobenius operator and expose escape rates and isolated resonances [Ulam, 1960, Keller, 1982, Froyland, 2007, 1995]. Agreement between the two routes is useful only when the domain, the hole, the orbit inclusion rule, and the finite-resolution error are held fixed. This requirement is especially important for reversible two-dimensional maps, where a visually stable leading eigenvalue can coexist with grid-phase oscillations.

We make this issue concrete for the area-preserving Hénon form

$$H_a(x, y) = (1 - ax^2 - y, x), \qquad a > 0, \tag{1}$$

which is conjugate to the usual $b = -1$ Hénon convention [Hénon, 1976, Wiggins, 1990]. The map has an exact reversor, reciprocal multiplier pairs, and an escape mechanism once a bounded open domain is fixed. Those structural facts are mathematically clean, but they do not make the whole plane uniformly hyperbolic: elliptic and escaping trajectories coexist with hyperbolic orbit families. We therefore separate exact statements on a certified survivor from numerical statements about a finite open operator.

The paper places exact symbolic structure, weighted periodic data, and finite-matrix spectral checks in one chain without assigning a continuous-operator meaning to finite-grid agreement. To separate direct coarse discretization from aggregation of a fixed fine estimator, we also use a paired algebraic projection. If P_M is a state-major matrix on a fine grid M and m divides M, the projected matrix is

$$C_{m\leftarrow M}(I,J) = r^{-2} \sum_{i\in B(I)} \sum_{j\in B(J)} P_M(i,j), \qquad r = M/m. \tag{2}$$

This creates a paired comparison with a direct coarse estimator. It changes both grid-specific sampling/layout and effective budget: a common row inherits sr^2 fine samples, whereas its direct control uses s. The comparison can therefore reveal sensitivity, but it cannot by itself identify a causal coupling contribution.

The main contributions are:

1. We prove that an explicit four-state-rectangle survivor at $a = 6$ is a locally maximal uniformly hyperbolic basic set conjugate to a mixing four-state subshift. This gives exact symbolic entropy and a primitive-necklace/orbit bijection on the local survivor.

2. We report the restricted weighted-cycle calculation through period twelve. The symbolic theorem makes the 79 primitive-orbit count and the $\beta = 0$ determinant identity exact; high-precision refinement supplies multipliers. We derive the dependence between the $\beta = 1$ Euler determinant and the periodic-point flat determinant, so their near-coincident finite-cutoff roots are used only as an implementation and truncation check.

3. We compare the period-twelve cycle reference with independently assembled open finite-volume matrices and report the leading values, eigenpair residuals, and returned spectral separation explicitly.

4. We test paired fine-to-coarse projection against direct coarse construction. The leading-modulus gaps are small at the tested resolutions, but dyadic-contrast compression is chain dependent. A proposed rowwise mechanism variable reduces to occupancy and supplies no robust internal-cell phase association after conditioning.

At the tested grids and sample budgets, the paired projection produces modest direct/projected leading-modulus gaps and compresses absolute dyadic contrast on one refinement chain. The inherited effective budget and the negative second chain prevent a causal or universal conclusion. The persisted row arrays likewise support only a target-occupancy restriction, not an internal-cell boundary-phase mechanism. These finite-resolution observations are kept separate from the exact survivor theorem: they do not identify a global Hénon zeta spectrum or establish a continuous-operator limit.

The remainder of the paper first defines the certified geometric object and the conditional zeta language, then describes the orbit and operator ledgers, and finally reports the direct and projected finite-resolution tests together with their limitations and reproducibility records.

2 Related work

Reversible maps and hyperbolic dynamics. The Hénon map introduced a low-dimensional testbed in which polynomial stretching, folding, and escape coexist [Hénon, 1976]. Reversibility and determinant-one Jacobian structure place the map near the broader theory of conservative and symplectic dynamics, but neither property implies uniform hyperbolicity [Katok and Hasselblatt, 1995, Wiggins, 1990]. Hyperbolic basic sets admit symbolic descriptions and periodic-orbit approximations under explicit hypotheses [Bowen, 1975, 1972, Lind and Marcus, 1995]. Our use of an explicit four-state-rectangle survivor follows this logic: the exact certificate is a local object, while the finite-volume dynamics on the corresponding open domain is treated only as a numerical diagnostic.

Closest Hénon work. Devaney and Nitecki established shift-automorphism descriptions for the Hénon mapping in a rigorous symbolic-dynamics setting [Devaney and Nitecki, 1979]. Covering relations provide another route to computer-assisted invariant sets in multidimensional maps [Zgliczyński and Gidea, 2004]. We do not present the first Hénon horseshoe, nor do we invoke a general covering-relation theorem. The narrower contribution here is an explicit set of rational state rectangles and constants, a direct transition-exclusion and signed-recurrence contraction certificate, and a separate finite-resolution operator comparison. These ingredients certify the stated local survivor and organize the numerical comparison without asserting a global Markov partition.

Dynamical zeta functions and cycle expansions. Artin and Mazur introduced a periodic-point zeta function for dynamical maps [Artin and Mazur, 1965]. Ruelle's dynamical determinants and thermodynamic formalism connect periodic data to transfer operators on expanding or Anosov systems [Ruelle, 1976, 1978], and Pollicott's work relates resonances to correlation decay [Pollicott, 1985]. The cycle expansion viewpoint organizes primitive orbits and stability weights into computable coefficients [Cvitanović et al., 2009, Gutzwiller, 1990]. We use these ideas as definitions and conditional comparisons. In particular, the finite-volume matrix assembled on four state rectangles is not automatically the same operator as a symbolic Ruelle operator, so equality is never inferred from a small leading-eigenvalue gap alone.

Finite-rank transfer-operator approximation. Ulam's finite-rank approximation is a standard way to turn a transfer operator into a stochastic matrix [Ulam, 1960, Keller, 1982]. Rigorous and numerical extensions cover invariant measures, isolated spectrum, metastability, and higher-dimensional maps [Froyland, 1995, 1998, 2007, Dellnitz and Junge, 1999, Bugiel, 1996, Bahsoun et al., 2015]. Spectral perturbation theory gives useful stability criteria when the underlying operator and the approximation satisfy compatible assumptions [Keller and Liverani, 1999, Liverani, 1995, Blank and Keller, 1998, 1997]. The present work focuses on a different failure mode: even when every finite matrix is nonnegative, substochastic, and has a small eigenpair residual, independent grid-level randomization can obscure the relationship between dyadic matrices. Our paired projection addresses this by retaining the same finest-grid cloud and inspecting rowwise differences.

Numerical spectral reliability. Finite matrices can contain spectral features that are sensitive to partition, sampling, or nonnormality. This motivates residual-based validation and independent reconstruction rather than visual eigenvalue comparison alone [Herwig et al., 2025, Nisoli, 2026]. Related work on coherent structures and isolated spectrum emphasizes that the relevant object is a joint property of the operator, function space, and approximation scheme [Froyland et al., 2010, 2011, Froyland and Dellnitz, 2003]. Our checker chain therefore persists matrix and schema hashes, rebuilds selected source rows independently, and reports the effective sample budget of a common-cloud projection.

Exploratory motivation. Interval-map symbolic dynamics and the author's earlier work on low-dimensional deterministic chaos motivate asking how weighted orbit data behave in a higher-dimensional reversible setting [Milnor and Thurston, 1988, Collet and Eckmann, 1980, Wang, 2026]. Periodic-orbit trace formulae also suggest longer-term semiclassical questions [Gutzwiller, 1990, Cvitanović et al., 2009]. The present comparison remains a dynamical-systems construction; number-theoretic or quantum interpretations would require additional operators, boundary conditions, and error estimates.

3 Geometry, reversibility, and the certified survivor

3.1 The reversible area-preserving map

We use the polynomial map

$$H_a(x, y) = (1 - ax^2 - y, x). \tag{3}$$

Its derivative and inverse are

$$DH_a(x, y) = \begin{pmatrix} -2ax & -1 \\ 1 & 0 \end{pmatrix}, \qquad H_a^{-1}(X, Y) = (Y, 1 - aY^2 - X). \tag{4}$$

Thus $\det DH_a = 1$. If $R(x, y) = (y, x)$, direct substitution gives

$$R \circ H_a \circ R = H_a^{-1}. \tag{5}$$

For a periodic point z_p of primitive period n_p, the monodromy matrix $M_p = DH_a^{n_p}(z_p)$ lies in $\mathrm{SL}(2, \mathbb{R})$. A hyperbolic orbit therefore has eigenvalues $(\Lambda_{u,p}, \Lambda_{u,p}^{-1})$ up to ordering, and the reversor maps it to an orbit with the same multiplier modulus. These are exact algebraic identities, not numerical observations.

The second-order recurrence associated with (3) is

$$x_{k+1} = 1 - ax_k^2 - x_{k-1}. \tag{6}$$

3.2 Closed state rectangles and the open numerical domain

The choice $a = 6$ is proof driven. Rearranging (6) gives two signed square-root branches, and the rational intervals below are mapped strictly into themselves on the six allowed neighbor patterns with the uniform contraction factor $2/\sqrt{17}$. They provide a simple explicit margin for a local symbolic model; no optimality in parameter or domain size is claimed. The rectangles were fixed before the matrix sensitivity comparisons reported below.

The exact survivor is defined with four closed state rectangles

$$N_{st} = X_s \times Y_t, \qquad s, t \in \{-, +\}, \tag{7}$$

with

$$X_- = [-5/8, -1/3], \quad X_+ = [1/3, 5/8], \qquad Y_- = [-81/128, -5/16], \quad Y_+ = [5/16, 81/128]. \tag{8}$$

We write $\mathcal{N} = \bigcup_{s,t} N_{st}$ and define the two-sided survivor

$$\Lambda_* = \bigcap_{k\in\mathbb{Z}} H_6^{-k}(\mathcal{N}). \tag{9}$$

The numerical operator instead uses the strict interior $\mathcal{N}^\circ = \bigcup_{s,t} \operatorname{int} N_{st}$ as its open domain: samples that hit a state-rectangle or local-cell boundary exactly are counted and discarded, and samples outside $\mathcal{N}^\circ$ contribute to escape. Historical filenames use the label “h-set,” but the present argument invokes neither the coordinate-chart structure of a topological h-set nor a covering relation in that technical sense.

The transition-exclusion and branch-contraction argument gives the state adjacency matrix, in the order $(--, -+, +-, ++)$,

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \rho(A) = \varphi = \frac{1+\sqrt{5}}{2}. \tag{10}$$

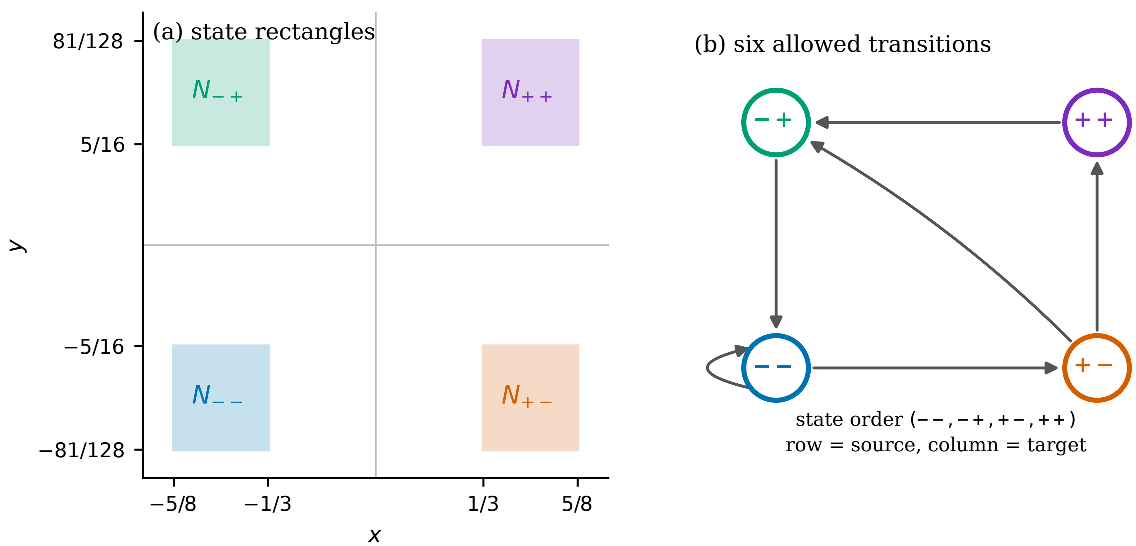

Figure 1: Geometry and symbolic coding. The left panel shows the four exact rational state rectangles in the (x, y) plane. The right panel shows the six allowed transitions in the order $(--, -+, +-, ++)$: $-- \to --, +-$, $-+ \to --$, $+- \to -+, ++$, and $++ \to -+$.

The following local statement is the theoretical anchor for the numerical ledgers.

Theorem 3.1 (Explicit hyperbolic state-rectangle survivor)**.** *The maximal invariant set Λ_* is nonempty, compact, invariant, and uniformly hyperbolic. For every bi-infinite itinerary admissible for A, the signed square-root recurrence induced by* (6) *has a unique orbit in the corresponding state rectangles. The survivor is conjugate to the four-state subshift defined by A, and*

$$h_{\text{top}}(H_6|_{\Lambda_*}) = \log\varphi. \tag{11}$$

Corollary 3.2 (A mixing hyperbolic basic set)**.** *The survivor Λ_* lies a positive distance inside $\mathcal{N}$, is locally maximal, and $H_6|_{\Lambda_*}$ is topologically mixing. In particular, Λ_* is a hyperbolic basic set.*

Proof. The strict self-mapping bounds proved in Lemma B.4, together with compactness, put Λ_* a positive distance from $\partial\mathcal{N}$. Choose an open neighborhood U with $\Lambda_* \subset U \subset \mathcal{N}$. Invariance gives $\Lambda_* \subset \bigcap_{k\in\mathbb{Z}} H_6^{-k}(U)$, while $U \subset \mathcal{N}$ gives the reverse inclusion by (9); hence Λ_* is locally maximal. Direct multiplication gives $A^4 > 0$, so the associated subshift of finite type is mixing. The conjugacy in Theorem 3.1 transfers mixing to $H_6|_{\Lambda_*}$. □

The theorem is local to the explicitly defined survivor. It does not identify the whole Hénon non-wandering set, certify a global Markov partition, or identify a finite-grid SCC with Λ_*. The exact signed-square-root map has Lipschitz constant at most $2/\sqrt{17} < 1$ on each admissible branch. The radicand bounds in the proof also give, uniformly along every survivor orbit,

$$\frac{\sqrt{17}}{12} \le |q_i| \le \sqrt{\frac{3}{8}}. \tag{12}$$

Both endpoints lie strictly between 1/3 and 5/8, and also strictly inside the corresponding y-interval magnitudes in (8); hence the positive rectangle margin used in Corollary 3.2 is explicit. The complete contraction, transition-exclusion, and cone proof is given in Appendix B; it uses neither the finite period-12 catalogue nor any finite-volume matrix.

Remark 3.3. The certified survivor and the finite-volume matrices answer different questions. The former is an exact local dynamical object. The latter are finite estimators with an absorbing escape rule. Comparing them is informative only after the domain and the estimator are explicitly recorded.

4 Weighted cycle expansions and error language

4.1 Definitions

Let $\mathcal{P}_*$ denote the primitive periodic orbits in the certified survivor Λ_*. For $p \in \mathcal{P}_*$, write n_p for its primitive period and $\Lambda_{u,p} = \sigma_p L_p$ for its unstable multiplier, where $\sigma_p \in \{-1, +1\}$ and $L_p = |\Lambda_{u,p}| > 1$. For $\beta \ge 0$ define the formal weighted dynamical zeta function and its Euler determinant by

$$Z_\beta(z) = \prod_{p\in\mathcal{P}_*} \left(1 - z^{n_p} L_p^{-\beta}\right)^{-1}, \qquad D_{E,\beta}(z) = Z_\beta(z)^{-1}. \tag{13}$$

Two different truncations occur and must be distinguished. The primitive-period cutoff retains all repetitions of orbits with $n_p \le N$:

$$\log Z_\beta^{(\mathrm{prim}\le N)}(z) = \sum_{\substack{p\in\mathcal{P}_* \\ n_p\le N}} \sum_{r\ge 1} \frac{z^{rn_p}}{rL_p^{\beta r}}. \tag{14}$$

The root calculations instead use the degree-N coefficient section

$$D_{E,\beta}^{[N]}(z) = \left[\prod_{\substack{p\in\mathcal{P}_* \\ n_p\le N}} \left(1 - z^{n_p} L_p^{-\beta}\right) \right]_{\le N}, \tag{15}$$

where $[\cdot]_{\le N}$ discards powers above z^N. Orbits of primitive period greater than N cannot affect these first N coefficients, but the polynomial in (15) is not the same object as the all-repetition expression in (14). At $\beta = 0$ this is the topological periodic-point zeta on the selected symbolic system; positive β weights unstable expansion. The notation does not imply that the finite catalogue is globally complete.

4.2 Periodic-point flat determinant and algebraic dependence

The second cycle polynomial is built from smooth fixed-point traces. If $M_p = DH_6^{n_p}(z_p)$, define

$$T_n = \sum_{x \in \mathrm{Fix}(H_6^n|_{\Lambda_*})} \frac{1}{|\det(I - DH_6^n(x))|} = \sum_{\substack{p \in \mathcal{P}_* \\ n_p | n}} \frac{n_p}{|\det(I - M_p^{n/n_p})|}. \tag{16}$$

The associated periodic-point flat determinant is the formal power series

$$D_F(z) = \exp\left(-\sum_{n \geq 1} \frac{T_n z^n}{n}\right). \tag{17}$$

Its degree-N coefficient section is computed from $f_0 = 1$ and

$$f_n = -\frac{1}{n} \sum_{j=1}^{n} T_j f_{n-j}, \qquad D_F^{[N]}(z) = \sum_{n=0}^{N} f_n z^n. \tag{18}$$

This definition is called “Fredholm” in the historical result fields, but no specific continuous operator or Banach space is inferred from that name here.

The Euler and flat determinants are not algebraically independent. Since the eigenvalues of M_p are $\sigma_p L_p$ and $\sigma_p L_p^{-1}$,

$$\frac{1}{|\det(I - M_p^r)|} = \sum_{k \geq 0} (k+1) \left(\sigma_p^k L_p^{-(k+1)}\right)^r. \tag{19}$$

Substitution into (17) gives the formal factorization

$$D_F(z) = \prod_{p \in \mathcal{P}_*} \prod_{k \geq 0} \left(1 - \sigma_p^k L_p^{-(k+1)} z^{n_p}\right)^{k+1}. \tag{20}$$

The $k = 0$ factor is exactly $D_{E,1}$. This formal algebraic dependence motivates comparing the leading roots of $D_{E,1}^{[N]}$ and $D_F^{[N]}$ as an empirical finite-section and implementation consistency check, not as independent cycle evidence. Degree truncation does not preserve the zero set of the full product, however, and the factorization supplies no bound on the finite-section root discrepancy.

4.3 Conditional operator statement

For a compact locally maximal uniformly hyperbolic basic set with a compatible Markov coding, standard thermodynamic formalism associates a weighted Ruelle operator to the potential

$$\phi_\beta = -\beta \log \|DH_a|_{E^u}\|. \tag{21}$$

Corollary 3.2 supplies the basic-set properties for Λ_*. Under the regularity assumptions of thermodynamic formalism, the corresponding dynamical determinant is related to the weighted periodic expansion [Ruelle, 1976, 1978, Bowen, 1975, Baladi, 2000, Blank et al., 2002, Baladi and Tsujii, 2016]. We use this statement only conditionally. The absorbing finite-volume matrix in Sections 6–6.1 is not identified with that symbolic operator by numerical proximity.

4.4 A finite-period tail bound

The following elementary bound records what can and cannot be inferred from a finite orbit table.

Lemma 4.1 (Conditional truncation bound)**.** *Assume that the number of primitive orbits of period n is at most Ce^{hn} with $h \geq 0$, and that $L_p \geq e^{\chi n_p}$ for all included orbits, with $\chi > 0$. If*

$$t = |z|e^{-\beta\chi} < 1, \qquad q = e^h t < 1,$$

then

$$\left|\log Z_\beta(z) - \log Z_\beta^{(\mathrm{prim}\leq N)}(z)\right| \leq \frac{Cq^{N+1}}{(1-q)(1-t^{N+1})}. \tag{22}$$

Proof. For a primitive orbit of period n, the geometric series in (14) is bounded by $t^n/(1-t^n)$. Summing over the omitted periods and using the assumed orbit count gives

$$\sum_{n>N} Ce^{hn}\frac{t^n}{1-t^n} \leq \frac{C}{1-t^{N+1}}\sum_{n>N} q^n,$$

which is (22). The assumptions, rather than the finite table, are what make the estimate valid. □

The lemma is useful as an accounting device for an absolutely convergent branch of the logarithm. It does not control a zero of a degree-N determinant section. In fact, the explicit bounds $h = \log\varphi$ and $\chi = \log(773/224)$ give the $\beta = 1$ domain $|z| < (773/224)/\varphi \approx 2.133$, while the reported reciprocal root is $z_* \approx 3.429$. The period-8–12 root behavior below is therefore empirical finite-section stabilization, not a consequence of Lemma 4.1.

4.5 What the zeta language excludes

The determinants above belong to the certified local survivor. The finite matrices below remain separately defined estimators; numerical proximity does not identify them with a continuous Ruelle operator or extend the determinant to the full Hénon plane.

5 Periodic-orbit catalogue and restricted operator construction

5.1 Catalogue bridge

The cycle ledger starts from the $a = 6$ periodic catalogue through period twelve. Each candidate was refined at high precision and classified relative to the four state rectangles in (7). The final classification contains 79 records inside the selected domain and 668 outside, with no unresolved or root-failed record. Theorem 3.1 and Lemma B.9 make the admissible symbolic-necklace count exact on Λ_*. In particular, Möbius inversion gives

$$p_n = \frac{1}{n}\sum_{d|n}\mu(d)\,\mathrm{tr}\left(A^{n/d}\right), \qquad (p_1,\ldots,p_{12}) = (1,0,1,2,2,2,4,5,8,11,18,25), \tag{23}$$

whose entries sum to 79. An independent checker re-enumerates the primitive words, reclassifies the persisted high-precision sequences, and verifies the trace/primitive-count identity. The numerical inside records agree period by period with the exact symbolic necklaces. The numerical catalogue therefore confirms the exact count while supplying coordinates and multipliers not contained in the symbolic enumeration.

For a periodic point z_p, the code stores the monodromy matrix M_p, its determinant residual, the unstable multiplier modulus, and the time-reversal partner when one exists. The reciprocal relation is checked numerically as a diagnostic of implementation quality; the exact determinant-one identity comes from (4).

5.2 Restricted weighted-cycle algebra and cutoff stability

On the certified symbolic survivor, the full unweighted Euler determinant is the exact finite-state identity

$$D_{E,0}(z) = Z_0(z)^{-1} = \prod_p (1 - z^{n_p}) = \det(I - zA) = 1 - z - z^3 - z^4, \tag{24}$$

where the product is over primitive symbolic necklaces. The stored degree-12 coefficient section reproduces this polynomial through degree twelve with zero error at the stored precision. For positive weights we form the truncated Euler and periodic-point flat polynomials from the same restricted orbit catalogue. For either polynomial, the reported leading modulus is $|z_*|^{-1}$, where z_* is its smallest-modulus nonzero root. The period-cutoff sequence in Table 1 shows stabilization of the $\beta = 1$ Euler and flat values near 0.29162845. The $\beta = 1/2$ Euler value near 0.68284718 is included only as a weight-sensitivity diagnostic; it is not the target of the finite-volume comparison.

Table 1: Restricted weighted-cycle cutoff sequence. The orbit count is cumulative through primitive period N.

N	primitive orbits	$\lambda_{E,1/2}$	$\lambda_{E,1}$	λ_F
8	17	0.682846853	0.291628304	0.291624276
9	25	0.682847077	0.291628328	0.291628474
10	36	0.682847204	0.291628476	0.291628512
11	54	0.682847184	0.291628450	0.291628441
12	79	0.682847185	0.291628451	0.291628452

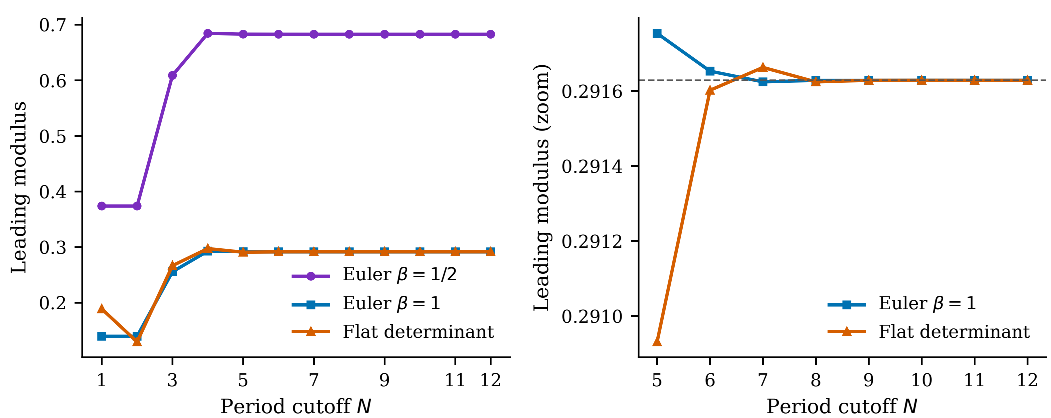

Figure 2: Restricted cycle-polynomial leading moduli versus period cutoff. The right panel resolves the agreement of the $\beta = 1$ Euler and periodic-point flat sections at the final cutoffs. Because the flat determinant contains the Euler determinant as its leading factor, this is a finite-section consistency check rather than independent dynamical evidence.

At $N = 12$, the two $\beta = 1$ diagnostics differ by 1.85×10^{-10} in leading modulus. In view of (20), their formal dependence motivates this empirical finite-truncation and implementation check. It neither implies equality of the degree-12 zero sets nor bounds their root discrepancy. The finite-volume comparison below uses their common displayed $\beta = 1$ value as a fixed cycle reference; it does not identify the finite matrix with a continuous thermodynamic operator.

5.3 Absorbing finite-volume matrices

Partition each state rectangle into an $m \times m$ array of cells B_I and write μ for planar Lebesgue measure. The ideal open finite-volume matrix is

$$(P_m)_{IJ} = \frac{\mu\Big(B_I \cap H_6^{-1}(B_J)\Big)}{\mu(B_I)}, \qquad P_m \in \mathbb{R}^{4m^2 \times 4m^2}. \tag{25}$$

Rows index source cells and columns index target cells. State-major ordering is followed by local y and local x indices. The implemented estimators replace the normalized cell integral in (25) by either tensor Gauss–Legendre weights, normalized to sum to one in each source cell, or equal-weight scrambled Sobol samples. A mapped point in B_J contributes to column J; a point outside $\mathcal{N}^\circ$, or exactly on a discarded state-rectangle or local-cell boundary, contributes to escape. The resulting CSR matrices are nonnegative and substochastic up to the recorded 10^{-12} audit tolerance.

The production panel uses $m \in \{24, 32, 48, 64, 96, 128\}$, Gauss–Legendre order eight, and two scrambled Sobol estimators with 64 points per cell. The independent schema checker verifies matrix hashes, indexing, selected source rows, zero exact boundary hits, and returned eigenpair residuals. Table 2 shows the six finest-grid comparisons with the period-12 cycle reference $\lambda_{\mathrm{cyc}} = 0.29162845164474605$.

Table 2: Finest-grid finite-volume comparison. The relative gap is $|\lambda_m - \lambda_{\mathrm{cyc}}|/|\lambda_{\mathrm{cyc}}|$.

Estimator	m	leading modulus λ_m	relative gap
Gauss $q = 8$	96	0.291998537	0.1269%
Gauss $q = 8$	128	0.288096349	1.2112%
Sobol 20260801	96	0.291078899	0.1884%
Sobol 20260801	128	0.292758121	0.3874%
Sobol 20260802	96	0.292764254	0.3895%
Sobol 20260802	128	0.290990842	0.2186%

The median relative gap is 0.3030%. The code applies ARPACK to the density transpose $P_m^\top$, normalizes each returned Ritz vector, and records

$$\eta_j = \left\| P_m^\top v_j - \lambda_j v_j \right\|_2, \qquad \|v_j\|_2 = 1. \tag{26}$$

Across these six records, all returned pairs have $\eta_j \le 1.66 \times 10^{-11}$, verifying small algebraic residuals for those Ritz pairs. The first/second modulus separation is 22.8%–29.2% of the first modulus, but only among the returned eigenvalues. Here "leading" means the largest modulus among the converged ARPACK eigenpairs requested with `which=LM`. For these nonnormal finite matrices, neither a small residual nor separation within the returned set proves proximity to a true eigenvalue or encloses the spectral radius. The historical dyadic stability gate remains negative because two Sobol trajectories are nonmonotone, so the table records a common finite-resolution window rather than convergence.

5.4 Variance and quadrature sensitivity

An extended panel adds sixteen Sobol seeds and paired budgets 64 and 256 on all six grids, together with Gauss orders 4, 8, and 12. Increasing the budget reduces cross-seed standard deviation at every grid; the six SD(256)/SD(64) ratios lie between 0.250 and 0.512, with median 0.3207. The strict paired-shift gate still fails at $m = 24$ and $m = 32$, and the q=8/q=12 comparison is not stable on all grids. Thus sampling variance is material but does not explain the whole historical failure.

The next section holds a finest-grid cloud fixed while changing the coarse representation. This supplies a paired sensitivity experiment rather than a replacement of the preceding estimators.

6 Operator cross-check and sensitivity

This section treats the finite-volume matrices as estimators, not as an already identified limiting operator. We report three layers of evidence: matrix integrity, direct/projected spectral comparison, and a rowwise mechanism test. Protocol identifiers and artifact hashes are deferred to Appendix A.

6.1 Fine-grid parents and common-cloud projection

The projection reuses matrices at $M = 96$ and $M = 128$ as fixed parents. Its fine-to-coarse direction is

$$96 \longrightarrow 48 \longrightarrow 24, \qquad 128 \longrightarrow 64 \longrightarrow 32. \tag{27}$$

Let $\mathcal{B}_M(I)$ be the block of fine-cell indices contained in coarse cell I, and define $\mathcal{B}_M(J)$ analogously for a coarse target cell. With $r = M/m$, the common-cloud matrix is

$$C_{m\leftarrow M}(I, J) = \frac{1}{r^2} \sum_{i\in\mathcal{B}_M(I)} \sum_{j\in\mathcal{B}_M(J)} P_M(i, j) = R_{m\leftarrow M} P_M E_{M\to m}. \tag{28}$$

Here

$$(R_{m\leftarrow M})_{Ii} = r^{-2}\mathbf{1}\{i \in \mathcal{B}_M(I)\}, \qquad (E_{M\to m})_{jJ} = \mathbf{1}\{j \in \mathcal{B}_M(J)\}, \tag{29}$$

with $R_{m\leftarrow M} \in \mathbb{R}^{4m^2\times 4M^2}$ and $E_{M\to m} \in \mathbb{R}^{4M^2\times 4m^2}$. Thus R averages the r^2 fine source rows in a coarse source block, while E sums fine target columns into their coarse target block. The producer stores the parent, direct-control, block-map, schema, and derived-matrix hashes. For a Sobol parent with s samples per fine cell, one projected coarse source row inherits sr^2 fine samples. Consequently, a direct/projected comparison changes both the sample layout and the effective finite estimator; it is not a matched-budget variance comparison. Below, "common-cloud" and "projected" refer to this same matrix C.

6.2 Direct/projected spectral gaps

For each projected matrix we compare the leading modulus λ_c with the corresponding direct-control modulus λ_d using

$$g = \frac{|\lambda_c - \lambda_d|}{\max(|\lambda_c|, |\lambda_d|)}. \tag{30}$$

Table 3 reports group medians over the 16 scramble seeds in each Sobol group. All eight medians are below the preregistered 2% descriptive threshold, and the pooled median over all 128 seed-level gaps is 0.0057803 (0.5780%), also below 2%. Thus the complete frozen gap component—at least six of eight group medians and the pooled median below 2%—passes. The seed distributions are heterogeneous, however: the largest gaps are 3.65% and 4.94% in two 64-sample coarse groups.

In total, 11 of the 128 seed-level gaps exceed 2%. Thus the result supports a shared group-level finite-resolution window, not a uniform per-seed bound. Because the projection also changes effective budget and aggregation geometry, it does not isolate a sampling-layout effect.

Table 3: Median direct/projected leading-modulus relative gaps ($n = 16$ seeds per group). Parent matrices are projected right-to-left, from $M = 96$ or 128 to the displayed targets; the chain label is written left-to-right only as the coarse-to-fine diagnostic reading order. Every group passes the 2% group-median threshold; individual seed values need not do so. Entries are fractions, so 0.007381 = 0.7381%.

Coarse-to-fine diagnostic chain	s	target m	median g
$24 \to 48 \to 96$	64	48	0.007381
$24 \to 48 \to 96$	64	24	0.015194
$24 \to 48 \to 96$	256	48	0.003279
$24 \to 48 \to 96$	256	24	0.003265
$32 \to 64 \to 128$	64	64	0.008301
$32 \to 64 \to 128$	64	32	0.012702
$32 \to 64 \to 128$	256	64	0.002620
$32 \to 64 \to 128$	256	32	0.002257

Means, maxima, seed fractions, and mean intervals are retained in Table 7.

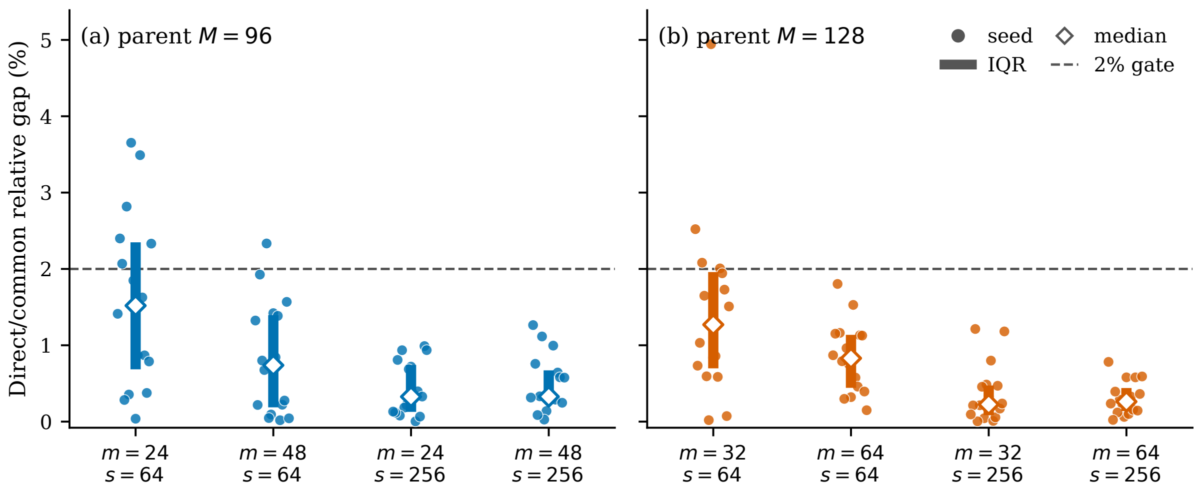

Figure 3: Direct/projected leading-modulus gaps for the eight Sobol groups. Points are the 16 scramble seeds; the summary marks the median and interquartile range. The dashed line is the 2% group-median decision threshold. All observations are retained, including the 11 of 128 values above the line.

6.3 Absolute dyadic-contrast compression

The matrices are projected from fine to coarse as in (27), but a three-level diagnostic trajectory is read in the opposite, coarse-to-fine direction. For $(\lambda_{\text{coarse}}, \lambda_{\text{middle}}, \lambda_{\text{fine}})$, define

$$\Delta_i = g(\lambda_{\text{coarse}}, \lambda_{\text{middle}}), \qquad \Delta_f = g(\lambda_{\text{middle}}, \lambda_{\text{fine}}), \qquad D = \Delta_f - \Delta_i. \tag{31}$$

Here g is the relative gap in (30). Negative D means that the final relative change is smaller than the initial one, but the preregistered criterion concerns absolute contrast. With direct and projected values D_d, D_c over the 16 seeds, support requires both

$$\text{median}\,|D_c| \le \text{median}\,|D_d|, \qquad \text{median}\left(\frac{|D_c|}{|D_d|}\right) \le 0.8. \tag{32}$$

The bootstrap unit is the scramble seed. Both budgets on the $24 \to 48 \to 96$ chain pass this absolute-contrast criterion, but only $s = 256$ has a negative projected mean D_c with its 95% interval below zero. Neither budget on the $32 \to 64 \to 128$ chain passes both inequalities. The frozen aggregate rule requires at least three of the four chain-by-budget groups to satisfy both inequalities in (32). Only two of four pass, so the aggregate dyadic component and the overall common-projection gate G1 both fail. The negative-D fractions and paired direct/projected differences are given in Table 8.

Table 4: Dyadic contrast D ($n = 16$). Table entries are fractions; multiplying by 100 gives the percentage-point units used in Figure 4. Chains are labeled in the coarse-to-fine diagnostic reading order; physical projection is from the rightmost parent to the left. Brackets give 95% seed-bootstrap intervals for the mean over 20,000 resamples. "Pass" refers exactly to (32), not to signed monotonicity.

Diagnostic chain	s	direct mean [CI]	projected mean [CI]	median ratio	gate
$24 \to 48 \to 96$	64	$-.00325$ [$-.01113$, .00497]	$+.00007$ [$-.00363$, .00459]	0.5630	pass
$24 \to 48 \to 96$	256	$-.00423$ [$-.00758$, $-.00067$]	$-.00294$ [$-.00407$, $-.00167$]	0.4616	pass
$32 \to 64 \to 128$	64	$-.00718$ [$-.01411$, $-.00110$]	$+.00477$ [.00236, .00730]	1.0521	fail
$32 \to 64 \to 128$	256	$-.00119$ [$-.00359$, .00095]	$+.00154$ [.00044, .00269]	0.8915	fail

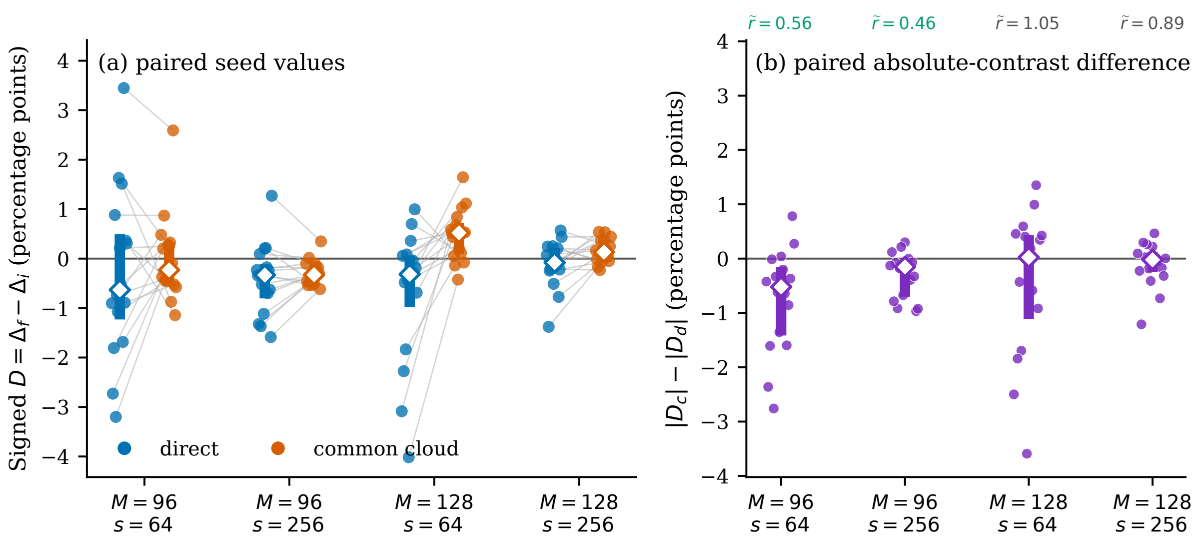

Figure 4: Seed-paired dyadic diagnostics. The left panel connects direct and projected signed contrasts $D = \Delta_f - \Delta_i$ for each of the 16 seeds. The right panel shows $|D_c| - |D_d|$, where negative values indicate absolute contrast compression; annotations give $\tilde{r} = \text{median}(|D_c|/|D_d|)$. Only the two $24 \to 48 \to 96$ groups satisfy the complete criterion (32).

The chain dependence prevents a universal compression claim. Because the projected rows also inherit larger effective budgets, even the positive first chain does not identify common sampling as the cause. The earlier negative dyadic records are retained as historical controls; no old decision is reopened.

6.4 Row support and a degenerate localization metric

For a direct row D_i and projected row C_i at the same target grid, define the row discrepancy energy

$$E_i = \frac{1}{2} \|D_i - C_i\|_1 . \tag{33}$$

The preregistered producer converts normalized sample distances into row-level exposure fractions at thresholds $\tau \in \{0.125, 0.25, 0.5, 1\}$. For samples inside a target state rectangle, the stored cell distance is

$$d_{\text{cell}} = \min(f_x, 1 - f_x, f_y, 1 - f_y), \qquad 0 \le d_{\text{cell}} \le \frac{1}{2}, \tag{34}$$

where f_x, f_y are fractional target-cell coordinates. More precisely, let $\mathcal{C}_i$ be the inherited parent sample cloud for coarse source row i, let w_ℓ be its quadrature or equal Sobol weight, and let I_ℓ indicate that the image lies inside a target state rectangle. The persisted statistic is

$$e_{\text{cell}}^{(\tau)}(i) = \begin{cases} \dfrac{\sum_{\ell \in \mathcal{C}_i} w_\ell I_\ell \mathbf{1}\{d_{\text{cell},\ell} \le \tau\}}{\sum_{\ell \in \mathcal{C}_i} w_\ell I_\ell}, & \sum\limits_{\ell \in \mathcal{C}_i} w_\ell I_\ell > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{35}$$

Thus the denominator is the inside-target weight, not the total parent-cloud weight, and a row with no inside target sample is assigned zero cell exposure.

At the preregistered $\tau = 1$ threshold, and also at $\tau = 0.5$, the bound $d_{\text{cell}} \le 1/2$ makes the cell-exposure array exactly

$$e_{\text{cell}}^{(\tau)}(i) = \mathbf{1}\{\text{inside fraction of row } i > 0\} \tag{36}$$

in all 136 persisted arrays. The preregistered localization variable therefore degenerates to occupancy: it is not an internal-cell boundary-phase test. Almost all discrepancy energy lies on occupancy-positive rows, with fraction 0.999864–1.000000 and mean 0.999987, but this support restriction is largely structural because a row with no target transition has little opportunity to differ.

To test whether any stronger signal survives, we condition on occupancy-positive Sobol rows and recompute lower-threshold associations. Every ρ is a rowwise Spearman correlation. Table 5 reports the mean of the 16 seed-level statistics in each group. Superscript + denotes this conditioning, and Q^+_τ is the energy fraction carried by the rank-defined top quartile of the conditional exposure array.

Table 5: Interpretation check for the cell-exposure metric. $24 \to 48 \to 96$ and $32 \to 64 \to 128$ are coarse-to-fine diagnostic labels; the parent grids are 96 and 128. ρ_{occ} is the formal $\tau = 1$ support-indicator correlation; lower-threshold columns condition on occupancy-positive rows. The rank-defined Q^+_τ values are tie-sensitive descriptive diagnostics, not inferential tests.

Diagnostic chain	s	m	ρ_{occ}	$\rho^+_{.125}$	$\rho^+_{.25}$	$Q^+_{.125}$	$Q^+_{.25}$
$24 \to 48 \to 96$	64	48	.9743	+.0889	+.0292	.2362	.2266
$24 \to 48 \to 96$	64	24	.9693	-.1990	-.3312	.1548	.1343
$24 \to 48 \to 96$	256	48	.9748	+.0922	+.0354	.2336	.2253
$24 \to 48 \to 96$	256	24	.9690	-.2067	-.3328	.1482	.1320
$32 \to 64 \to 128$	64	64	.9766	+.0875	+.0199	.2411	.2313
$32 \to 64 \to 128$	64	32	.9726	+.0829	-.0034	.2200	.2055
$32 \to 64 \to 128$	256	64	.9768	+.0922	+.0199	.2394	.2295
$32 \to 64 \to 128$	256	32	.9722	+.1000	+.0065	.2203	.2051

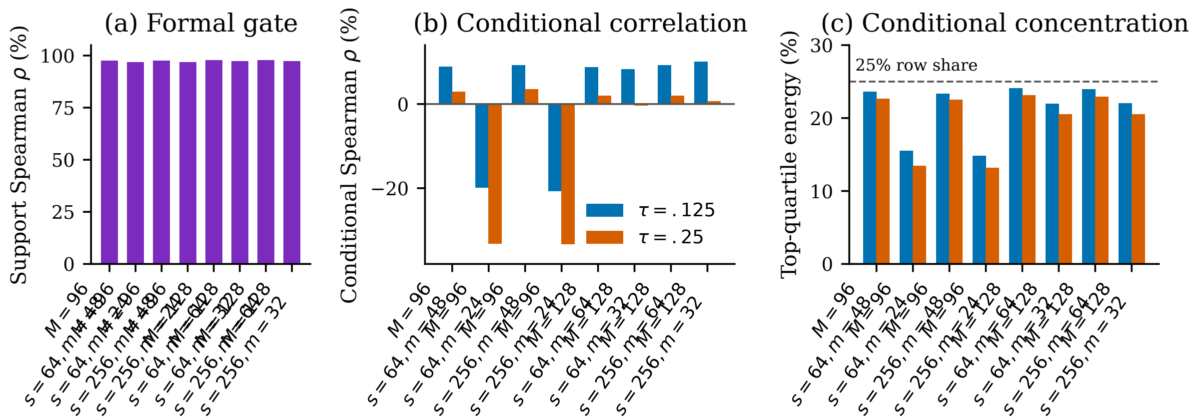

Figure 5: Localization-metric diagnosis. The formal $\tau = 1$ statistic is a binary target-support indicator. Axis labels give parent grid M, sample budget s, and target grid m; bars summarize the 16 seed-level statistics per group. After conditioning on support-positive rows, the $\tau = 0.125$ and 0.25 cell-boundary correlations are small or negative, and the corresponding top-quartile energy fractions lie near or below the 25% row-share reference.

The conditional correlations range from -0.333 to 0.100 across the eight groups and provide no robust positive boundary-phase association. Thus the metric fails as a mechanism variable: the only surviving statement is a finite-row support restriction, while internal target-cell phase remains unresolved. This reinterpretation changes no persisted array or recorded gate value.

6.5 Quadrature control and scope

On the projected Gauss matrices, the $q = 8$ versus $q = 12$ leading-modulus gaps are 0.1054%, 0.0680%, 0.0712%, and 0.0881% for target grids 24, 32, 48, and 64. Thus all four are below the 1% control threshold. This controls quadrature-order sensitivity for these finite matrices; it neither repairs the failed localization metric nor proves an $m \to \infty$ operator limit. Exact commands, parent hashes, independent checkers, and collection digests are recorded in Appendix A.

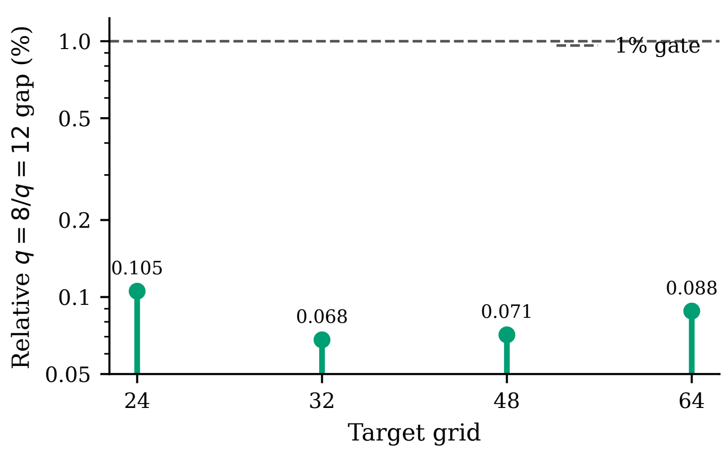

Figure 6: Projected Gauss $q = 8$ versus $q = 12$ leading-modulus gaps at the four target grids. A logarithmic vertical scale displays both the observed 0.068%–0.105% range and the 1% decision line.

7 Discussion, limitations, and outlook

The exact and numerical parts of the paper answer complementary questions. The state-rectangle theorem identifies a local mixing hyperbolic basic set and turns symbolic necklaces into exact survivor-orbit counts. The high-precision catalogue then supplies the multiplier data needed for weighted cycle sections. Separately assembled finite-volume matrices occupy the same finite-resolution leading-modulus window at the finest tested grids, but their nonmonotone dyadic trajectories show that this agreement is not itself a convergence theorem.

The paired projection experiment sharpens this distinction. All eight direct/common-cloud group medians and their pooled median are below 2%, yet some individual 64-sample seeds reach 3.65% and 4.94%. Absolute dyadic-contrast compression passes only two of the four chain-by-budget groups, short of the frozen three-group aggregate requirement; hence overall G1 fails. Only the $s = 256$ group on the $24 \to 48 \to 96$ diagnostic chain has a projected mean contrast with a wholly negative interval. These observations reveal a reproducible sensitivity split, while the inherited sr^2 projection budget prevents attribution to common sampling alone. The rowwise analysis is similarly cautionary: its preregistered localization variable equals target occupancy at the decisive threshold, and conditioning on occupied rows leaves no robust internal-cell phase association.

7.1 Limitations

The map acts on a noncompact plane containing elliptic, hyperbolic, and escaping trajectories. The exact theorem applies only to the explicitly defined survivor Λ_*. The finite-volume matrices use a strict absorbing hole and a state-major rectangular partition; they are not proved to converge to a specified continuous operator. The residuals in (26) verify small algebraic residuals for the returned Ritz pairs; for a nonnormal matrix they do not prove eigenvalue proximity or that ARPACK found the global spectral radius. Likewise, the period-8–12 root stability is empirical: the tail estimate in Lemma 4.1 does not reach the reported reciprocal root.

The projected and direct coarse estimators differ in effective sample budget and layout. Their comparison therefore cannot isolate a causal common-random-number effect. The row statistics are computed over finite sample clouds, and the original localization metric degenerates to support. A geometric statement about image-boundary regularity would require a new occupancy-conditioned metric and an error analysis on a common function space.

7.2 Code and data availability

The R059–R061 protocols, JSON/CSV ledgers, CSR matrices, compact row arrays, independent checkers, and SHA-256 manifests are organized in the accompanying artifact tree. Exact commands and final collection hashes are recorded in `research/refine-logs/R059_CERTIFIED_DOMAIN_MANIFEST.md`, `research/refine-logs/R060_OPERATOR_VARIANCE_MANIFEST.md`, and `research/refine-logs/R061_COMMON_CLOUD_MANIFEST.md`, and are summarized in Appendix A. The production machine provided 32 AMD EPYC vCPUs and 60 GB of memory; at most eight workers were used in the final projection run. A public archival URL and immutable release identifier have not yet been assigned and must be added before submission; no placeholder address is asserted here.

7.3 Outlook

The most informative next experiment is a matched-budget, common-cloud design with an independent direct control and preregistered neighboring grid phases. For the matrix problem, positive-vector Collatz–Wielandt bounds or an independent spectral-radius enclosure would strengthen the meaning of the reported leading value. For the cycle problem, an isolating contour and a Rouché-type remainder bound could turn the observed root stabilization into a certificate. The explicit basic set also offers a clean setting for exploring alternative Hölder potentials, signed multiplier weights, and eventually a specified semiclassical quantization. These directions are deliberately exploratory: the present construction supplies a concrete testbed on which stronger statements can be formulated and checked.

7.4 Conclusion

An explicit local Hénon survivor can support both exact symbolic dynamics and a transparent finite-resolution comparison between weighted cycles and open finite-volume matrices. The strongest robust conclusion is mixed but useful: the six tested parent-grid values lie within 1.22% of the period-12 cycle reference, while all eight direct/common-cloud group medians lie below 2%. The aggregate dyadic gate nevertheless fails, and the proposed localization mechanism does not survive conditioning. This separation of theorem, finite-section evidence, and unresolved mechanism provides a controlled base for sharper operator and determinant certificates.

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A Supplementary audit ledger

A.1 R059–R061 gate summary

Table 6: Recorded gate decisions preserved with the artifact ledger.

Run	Gate	Decision
R059	matrix integrity and cross-method controls	pass
R059	dyadic stability G4	**fail; retained**
R060	matrix integrity G0	pass
R060	Sobol paired-shift G1	fail
R060	Gauss-order G2	fail
R060	ensemble-mean G3	pass
R061	integrity G0	pass
R061	common/direct gap component	pass
R061	absolute dyadic-contrast component	**fail; mixed**
R061	common-projection G1 overall	**fail; mixed**
R061	row localization G2	formal pass; interpretation downgraded
R061	q=8/q=12 leading-modulus control G3	pass

A.2 R060 frozen-count text inconsistency

The frozen R060 G0 prose says "162 configurations," but the same protocol's Cartesian product is $6[2\times16+3] = 210$. The producer and independent checker use the design-derived 210 configurations (192 Sobol and 18 Gauss) and record the text inconsistency explicitly. The protocol was not edited after freezing.

A.3 R061 protocol counts and hashes

The frozen R061 protocol has SHA-256:

```
64c48efb34f4fbf82164148a8b10b4d9
66f6f01a3d8a6fa7fb57cccffa647178
```

The production contains 68 read-only R060 parent references, 136 derived CSR matrices, and 136 direct-control links. The two independent checkers return `all_checks_pass=true`. The matrix collection contains 272 files (136 NPZ files and 136 schema files), and the row-localization collection contains 136 NPZ files. Exact file hashes and collection digests are recorded in the manifest cited in Section 7.

A.4 Effective sample accounting

If the fine grid is M and the target grid is m, then $r = M/m$ and a coarse source row aggregates r^2 fine source cells. The common-cloud estimator thus inherits sr^2 fine samples per coarse source cell. This accounting is important for interpreting the direct/common comparison:

$$s_{\text{eff}} = sr^2, \qquad (s, M, m) \in \{(64, 96, 48), (64, 96, 24), (64, 128, 64), (64, 128, 32), \dots\}. \tag{37}$$

For $r = 2$ and $r = 4$, the effective budgets are therefore four and sixteen times the direct budget. No common-cloud record is pooled with a direct record as if both had the same sampling budget, and no causal coupling effect is estimated from this confounded comparison.

A.5 Seed-level direct/projected gap summaries

Table 7 retains the heterogeneity hidden by a median-only summary. Chain A denotes $24 \to 48 \to 96$ and chain B denotes $32 \to 64 \to 128$ in coarse-to-fine diagnostic order; their physical parent grids are $M = 96$ and $M = 128$, respectively, and projection runs fine to coarse. The interval is the conventional 95% Student-t interval for the seed mean; it is a read-only descriptive summary of the 16 persisted seed values. The final column is the fraction of individual seeds with $g \leq 2\%$; the preregistered group decision itself used only the median.

Table 7: Full direct/projected relative-gap summaries, in percent.

Chain	s	m	n	mean [95% CI]	median	maximum	seed fraction $\leq 2\%$
A	64	48	16	0.8247 $[0.4255, 1.2239]$	0.7381	2.3335	0.9375
A	64	24	16	1.5765 $[0.9658, 2.1872]$	1.5194	3.6527	0.6250
A	256	48	16	0.5009 $[0.3031, 0.6987]$	0.3279	1.2644	1.0000
A	256	24	16	0.4328 $[0.2446, 0.6209]$	0.3265	0.9896	1.0000
B	64	64	16	0.8425 $[0.5945, 1.0906]$	0.8301	1.8035	1.0000
B	64	32	16	1.4447 $[0.8103, 2.0792]$	1.2702	4.9438	0.7500
B	256	64	16	0.3114 $[0.1927, 0.4302]$	0.2620	0.7823	1.0000
B	256	32	16	0.3736 $[0.1678, 0.5794]$	0.2257	1.2134	1.0000

A.6 Full dyadic-contrast decision fields

The direct/projected negative fractions in Table 8 are the proportions of seeds with $D \leq 0$. The paired-difference interval is a descriptive Student-t interval for $D_c - D_d$ and was not part of the frozen decision rule. Chain labels retain the same coarse-to-fine diagnostic convention as in Table 7.

Table 8: Seed-level dyadic decision fields ($n = 16$ in every row).

Chain	s	$\Pr(D_d \leq 0)$	$\Pr(D_c \leq 0)$	median $\lvert D_c\rvert/\lvert D_d\rvert$	mean $D_c - D_d$ [95% CI]	absolute gate
A	64	0.5625	0.5625	0.5630	0.00333 $[-0.00453, 0.01118]$	pass
A	256	0.7500	0.8750	0.4616	0.00129 $[-0.00160, 0.00418]$	pass
B	64	0.6875	0.1875	1.0521	0.01195 $[0.00484, 0.01906]$	fail
B	256	0.6250	0.3125	0.8915	0.00273 $[0.00032, 0.00514]$	fail

A.7 Localization interpretation check

The read-only check `results/localization_interpretation_audit_r061.json` has SHA-256

`bd7c4b9fb8d737783ef95aa5054fa9c6 7d7f7d011a33f442a4f331aefea09ea3.`

All 136 persisted array hashes match. Because the stored internal-cell distance is at most one half of a cell width, the $\tau = 1$ and $\tau = 0.5$ cell-exposure arrays exactly equal the indicator that a row has nonzero target occupancy. The formal G2 pass is therefore retained only as an occupancy/support association. Conditional on occupancy-positive rows, the eight Sobol-group mean correlations range from -0.207 to 0.100 at $\tau = 0.125$ and from -0.333 to 0.035 at $\tau = 0.25$; no robust positive internal-cell phase conclusion is made.

A.8 Scope boundary

The manuscript uses "finite-resolution evidence" for all R060/R061 operator statements. It does not assert an $m \to \infty$ limit, continuous transfer-operator convergence, a global Hénon zeta, a Riemann-zero correspondence, a Hilbert–Pólya construction, or evidence for the Riemann hypothesis.

B Proof of the explicit state-rectangle theorem

This appendix gives the proof used by Theorem 3.1. It is included to make the logical dependency of the numerical ledgers explicit. Starting from the exact rational state-rectangle intervals in (8) and the matrix A in (10), we prove the one-step exclusion, signed-recurrence contraction, conjugacy, period preservation, entropy identity, and two-sided cone bounds. No finite period catalogue or finite-volume matrix enters the proof.

B.1 State paths and the signed recurrence

Let Σ_A be the space of bi-infinite state sequences $\omega = (w_i)_{i\in\mathbb{Z}}$ with $A_{w_i w_{i+1}} = 1$, equipped with the product topology, and let σ denote the left shift. Write $w_i = (s_i, t_i)$ with $s_i, t_i \in \{-, +\}$. The nonzero entries of A imply

$$t_i = s_{i-1} \qquad (i \in \mathbb{Z}). \tag{38}$$

We henceforth identify the symbols $+$ and $-$ with the numbers $+1$ and -1, respectively. Thus an admissible state path has a unique scalar sign sequence $\varepsilon_i = s_i \in \{-1, +1\}$ and $w_i = (\varepsilon_i, \varepsilon_{i-1})$. The only additional restriction is that the two neighbors of a central symbol are not both positive:

Lemma B.1 (Exact rectangle separation and one-step exclusion)**.** *The four rectangles are pairwise disjoint, X_s lies strictly inside Y_s for each sign, and every orbit in Λ_* has an A-admissible state itinerary.*

Proof. The endpoint ordering

$$\frac{5}{16} < \frac{1}{3} < \frac{5}{8} < \frac{81}{128}$$

and its reflection prove separation and the strict inclusions $X_s \subset \operatorname{int} Y_s$. If $(x, y) \in N_{st}$ and $H_6(x, y) \in N_{s't'}$, then the second image coordinate is x; hence $t' = s$. This excludes eight of the ten zero entries of A. For the two remaining structural pairs, the source has $y \in Y_+$ and the target has x-sign positive. On either such source,

$$1 - 6x^2 - y \le 1 - 6\left(\frac{1}{3}\right)^2 - \frac{5}{16} = \frac{1}{48} < \frac{1}{3} = \inf X_+.$$

Thus these two transitions are also impossible, leaving exactly the six nonzero entries of A. $\square$

Lemma B.2 (Graph/sign equivalence)**.** *For a cyclic or bi-infinite path, an edge $(\varepsilon_i, \varepsilon_{i-1}) \to (\varepsilon_{i+1}, \varepsilon_i)$ is A-allowed if and only if ε_{i-1} and ε_{i+1} are not both $+$.*

Proof. The overlap condition (38) leaves two possible targets for each source state. In the state order $(--, -+, +-, ++)$ these structural possibilities are

$$-- \to --, +-, \qquad -+ \to --, +-, \qquad +- \to -+, ++, \qquad ++ \to -+, ++\,.$$

The matrix A deletes exactly $-+ \to +-$ and $++ \to ++$. These are precisely the two cases in which the predecessor and successor signs are both positive; all other structural edges are retained. $\square$

For a sign sequence ε, define the closed sign box

$$K_\varepsilon = \prod_{i\in I} X_{\varepsilon_i},$$

where $I = \mathbb{Z}$ for a bi-infinite path and $I = \mathbb{Z}/n\mathbb{Z}$ for a cyclic word. On this box consider

$$(T_\varepsilon q)_i = \varepsilon_i \sqrt{\frac{1 - q_{i-1} - q_{i+1}}{6}}. \tag{39}$$

Lemma B.3 (Rational radicand bounds)**.** *If q_{i-1}, q_{i+1} lie in their prescribed sign intervals and are not both positive, then*

$$\frac{1 - q_{i-1} - q_{i+1}}{6} \in \left[\frac{5}{18}, \frac{3}{8}\right] \quad \text{or} \quad \left[\frac{17}{144}, \frac{31}{144}\right].$$

In either case its positive square root lies strictly between $1/3$ and $5/8$.

Proof. If both neighbors are negative, their sum belongs to $[-5/4, -2/3]$, giving $[5/18, 3/8]$ after the affine rescaling. If one is positive and one negative, their sum belongs to $[-7/24, 7/24]$, giving $[17/144, 31/144]$. Finally,

$$5/18 > 1/9, \quad 3/8 < 25/64, \quad 17/144 > 1/9, \quad 31/144 < 25/64,$$

so taking positive square roots gives the claimed strict bounds. □

Lemma B.4 (Strict self-mapping)**.** *For every admissible sign sequence, T_ε maps K_ε into its coordinatewise interior.*

Proof. Lemma B.2 excludes the only case in which both neighbors are positive. Lemma B.3 then gives $1/3 < \sqrt{(1 - q_{i-1} - q_{i+1})/6} < 5/8$. Multiplication by ε_i places the result in the interior of X_{ε_i} at every coordinate. □

Lemma B.5 (Uniform contraction)**.** *On every admissible sign box,*

$$\|T_\varepsilon q - T_\varepsilon q'\|_\infty \le \frac{2}{\sqrt{17}} \|q - q'\|_\infty < \|q - q'\|_\infty .$$

Proof. For $f(u, v) = \sqrt{(1 - u - v)/6}$, the derivative bound on the intervals of Lemma B.3 is

$$\left|\frac{\partial f}{\partial u}\right| = \frac{1}{12\sqrt{(1 - u - v)/6}} \le \frac{1}{\sqrt{17}},$$

and the same holds for v. The mean-value theorem therefore gives

$$|(T_\varepsilon q)_i - (T_\varepsilon q')_i| \le \frac{1}{\sqrt{17}}(|q_{i-1} - q'_{i-1}| + |q_{i+1} - q'_{i+1}|) \le \frac{2}{\sqrt{17}} \|q - q'\|_\infty .$$

For cyclic words of length one or two, the two neighbor occurrences may refer to the same coordinate; their derivative contributions add and still give the same coefficient $2/\sqrt{17}$. Since $4 < 17$, this coefficient is strictly less than one. □

Lemma B.6 (One fixed sequence per admissible itinerary)**.** *Every finite cyclic or bi-infinite admissible sign sequence has exactly one fixed point of T_ε in K_ε.*

Proof. For a cyclic word, K_ε is a finite product of closed intervals and is therefore complete. For a bi-infinite word, it is a closed subset of $\ell^\infty(\mathbb{Z})$ and is complete in the sup norm. This norm topology on each fixed sign box is distinct from the product topology used on the symbolic space Σ_A below. Lemma B.4 gives self-mapping, and Lemma B.5 gives a contraction with constant less than one. Banach's fixed-point theorem applies. □

B.2 Orbit correspondence and conjugacy

Lemma B.7 (Fixed sequences are Hénon orbits)**.** *If $q = T_\varepsilon q$, then*

$$q_{i+1} = 1 - 6q_i^2 - q_{i-1}, \qquad z_i = (q_i, q_{i-1}) \in N_{\varepsilon_i,\varepsilon_{i-1}},$$

and $H_6(z_i) = z_{i+1}$.

Proof. Squaring the fixed-point equation gives $6q_i^2 = 1 - q_{i-1} - q_{i+1}$, which is the recurrence. The strict self-mapping places q_i in X_{ε_i}, while $X_{\varepsilon_{i-1}}$ lies in the interior of $Y_{\varepsilon_{i-1}}$. Substitution into the definition of H_6 gives the map identity. □

Lemma B.8 (Conjugacy on the full state-rectangle survivor)**.** *The itinerary map $\pi : \Lambda_* \to \Sigma_A$ is a homeomorphism and*

$$\pi \circ H_6 = \sigma \circ \pi.$$

Proof. Every set $H_6^{-k}(\mathcal{N})$ is closed, and the intersection in (9) is contained in the compact set $\mathcal{N}$ supplied by the $k = 0$ term. Hence Λ_* is compact. Shifting the integer index in (9) proves invariance. The constant $--$ itinerary is admissible, so Lemmas B.6 and B.7 show that Λ_* is nonempty.

For $\omega \in \Sigma_A$, let $q(\omega)$ be the unique fixed sequence from Lemma B.6, and define

$$\Phi(\omega) = \big(q_0(\omega), q_{-1}(\omega)\big).$$

Lemma B.7 places the full orbit of this point in $\mathcal{N}$, so $\Phi(\omega) \in \Lambda_*$. We next prove that Φ is continuous without comparing two different branch maps globally. Suppose that two sign sequences agree on $[j - m, j + m]$, and let q, q' be their fixed sequences. For $0 \le r \le m$, set

$$D_r = \max_{|i-j| \le m-r} |q_i - q_i'|.$$

The common sign at every coordinate in the initial window gives $D_0 \le 7/24$, the diameter of either interval X_s. At each coordinate in the smaller window defining D_{r+1}, both neighboring coordinates lie in the window defining D_r, and the two fixed-point equations use the same central sign. The derivative estimate from Lemma B.5 therefore gives

$$D_{r+1} \le \frac{2}{\sqrt{17}} D_r, \qquad |q_j - q_j'| \le \frac{7}{24} \left(\frac{2}{\sqrt{17}}\right)^m.$$

If two itineraries agree on $[-m, m]$, this bound applies at $j = 0$ with depth m and at $j = -1$ with depth $m - 1$. Thus both coordinates of Φ vary continuously in the product topology.

Conversely, every $z \in \Lambda_*$ has a unique state at every iterate because the four state rectangles are pairwise disjoint. Lemma B.1 makes this itinerary A-admissible and defines $\pi(z) \in \Sigma_A$. Writing $H_6^i(z) = (q_i, q_{i-1})$, the Hénon recurrence and the sign of q_i recover the signed square-root equation $q = T_\varepsilon q$. Uniqueness in Lemma B.6 then gives $\Phi \circ \pi = \mathrm{id}_{\Lambda_*}$, while construction gives $\pi \circ \Phi = \mathrm{id}_{\Sigma_A}$. The subshift Σ_A is compact in the product topology and Λ_* is Hausdorff, so the continuous bijection Φ is a homeomorphism. Finally, $\Phi \circ \sigma = H_6 \circ \Phi$, equivalently $\pi \circ H_6 = \sigma \circ \pi$. □

Call the shift orbit of a periodic A-admissible state sequence a *symbolic necklace*; it is primitive when its least shift period equals its length. Thus rotations of a based cyclic word represent the same necklace.

Lemma B.9 (Period preservation and orbit bijection)**.** *The conjugacy induces a bijection between primitive symbolic necklaces of length n and primitive H_6-orbits in Λ_* of period n.*

Proof. For a based cyclic word of least period n, the corresponding scalar sequence cannot have a smaller period $d < n$, because its signs would then also have period d. The state itinerary has the same least period because its second sign is the previous first sign. Rotating the based word shifts the fixed sequence and hence selects another point on the same H_6-orbit. Conversely, the itinerary of every periodic orbit is periodic, and the conjugacy preserves least periods. Quotienting both sides by these shifts gives the stated bijection. $\square$

B.3 Entropy and scope

Lemma B.8 proves the conjugacy and Lemma B.9 gives the primitive periodic-orbit bijection. The entropy of a finite-type subshift is the logarithm of the spectral radius of its adjacency matrix. Direct calculation gives

$$\det(\lambda I - A) = (\lambda^2 - \lambda - 1)(\lambda^2 + 1),$$

so $\rho(A) = (1 + \sqrt{5})/2$ and $h_{\mathrm{top}}(H_6|_{\Lambda_*}) = \log \rho(A)$.

B.4 Two-sided cone certificate

For completeness, we record the exact hyperbolicity bounds on the same four state rectangles. Their common horizontal and vertical half-widths are

$$\alpha = \frac{7}{48}, \qquad \beta = \frac{41}{256}, \qquad r = \frac{\beta}{\alpha} = \frac{123}{112}.$$

In normalized tangent coordinates $(u, v) = (\delta x/\alpha, \delta y/\beta)$, define

$$\mathcal{C}^u = \{(u, v) : |v| \le |u|/2\}, \qquad \mathcal{C}^s = \{(u, v) : |u| \le |v|/2\}.$$

For $z = (x, y)$ in the state-rectangle union,

$$\widehat{DH_6}_z = \begin{pmatrix} -12x & -r \\ r^{-1} & 0 \end{pmatrix}.$$

If $(u, v) \in \mathcal{C}^u$, then $|x| \ge 1/3$ and

$$|u'| \ge \left(4 - \frac{123}{224}\right)|u| = \frac{773}{224}|u|, \qquad \frac{|v'|}{|u'|} \le \frac{25088}{95079} < \frac{1}{2}.$$

Thus the unstable cone is mapped strictly into itself and expands by at least $773/224$ in the normalized max norm. Since $H_6^{-1}(X, Y) = (Y, 1 - 6Y^2 - X)$,

$$\widehat{DH_6^{-1}}_Z = \begin{pmatrix} 0 & r \\ -r^{-1} & -12Y \end{pmatrix}.$$

For $(u, v) \in \mathcal{C}^s$ and $|Y| \ge 5/16$,

$$|v'| \ge \left(\frac{15}{4} - \frac{56}{123}\right)|v| = \frac{1621}{492}|v|, \qquad \frac{|u'|}{|v'|} \le \frac{15129}{45388} < \frac{1}{2}.$$

Hence the stable cone is strictly invariant under DH_6^{-1} with backward expansion at least $1621/492$.

To see that these pointwise inequalities yield invariant lines, write an unstable slope as $m = v/u$. Its projective map is

$$T_x(m) = \frac{r^{-1}}{-12x - rm}, \qquad |T_x'(m)| \le \left(\frac{224}{773}\right)^2 < 1.$$

The corresponding stable slope $n = u/v$ under DH_6^{-1} satisfies

$$S_Y(n) = \frac{r}{-r^{-1}n - 12Y}, \qquad |S_Y'(n)| \le \left(\frac{492}{1621}\right)^2 < 1.$$

We spell out the graph-transform limit. Write $z_k = H_6^k(z) = (x_k, y_k)$ and let $I = [-1/2, 1/2]$. For the unstable direction, define

$$F_{n,z}^u = T_{x_{-1}} \circ T_{x_{-2}} \circ \cdots \circ T_{x_{-n}}.$$

Every factor maps I strictly into I and has Lipschitz constant at most $\kappa_u = (224/773)^2$. Hence $F_{n,z}^u(0)$ is uniformly Cauchy: after one more factor is appended in the remote past, the change at time zero is at most $\kappa_u^n \operatorname{diam}(I)$. The limit is independent of the initial slope for the same reason. Each finite composition depends continuously on z, and uniform convergence makes the limiting slope $m^u(z)$ continuous. Removing the most recent factor gives

$$m^u(H_6 z) = T_{x_0}(m^u(z)),$$

which is precisely invariance of the line E_z^u under DH_6.

For the stable direction, use the future orbit and

$$F_{n,z}^s = S_{y_1} \circ S_{y_2} \circ \cdots \circ S_{y_n}.$$

The identical argument with $\kappa_s = (492/1621)^2$ gives a continuous limit $n^s(z)$, independent of its terminal slope. Shifting the future composition shows that DH_6^{-1} maps the resulting line at z_k to the line at z_{k-1}, equivalently $DH_6 E_{z_{k-1}}^s = E_{z_k}^s$. We have therefore constructed continuous invariant lines $E_z^u \subset \mathcal{C}^u$ and $E_z^s \subset \mathcal{C}^s$.

The cones intersect only at zero, so $T_z\mathbb{R}^2 = E_z^u \oplus E_z^s$; the expansion bounds above give uniform forward expansion on E^u and uniform forward contraction on E^s. Thus Λ_* is uniformly hyperbolic. Together with the preceding lemmas, this completes the proof of Theorem 3.1.

The result is deliberately local. It does not certify the whole Hénon non-wandering set, a global Markov partition, or a finite-grid strongly connected component, and it does not identify the restricted continuous operator with any finite-volume matrix or with a global Hénon zeta.